as viewed from a distant vantage point—were placed inside such a rotating sphere, the calculations show that the spinning space would exert a force on the stationary water, causing it to rise up the bucket walls and take on a concave shape.

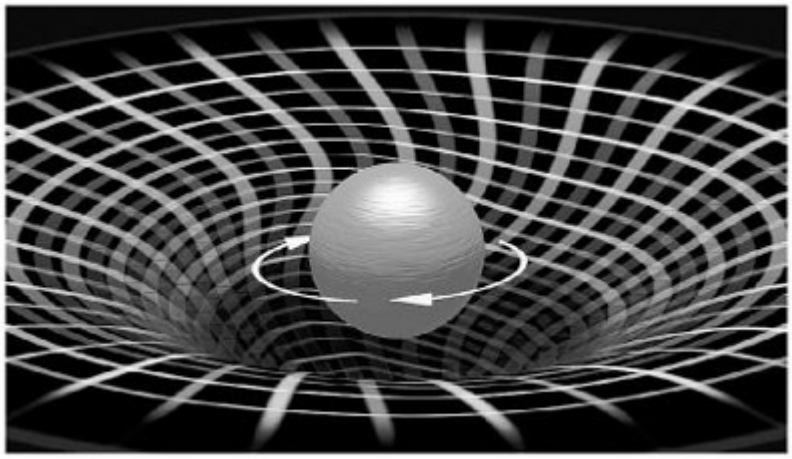


Figure 14.1 A massive spinning object drags space—the freely falling frames—around with it.

This result would have pleased Mach no end. Although he might not have liked the description in terms of “spinning space”—since this phrase portrays spacetime as a something—he would have found it extremely gratifying that *relative* spinning motion between the sphere and the bucket causes the water’s shape to change. In fact, for a shell that contains enough mass, an amount on a par with that contained in the entire universe, the calculations show that it doesn’t matter one bit whether you think the hollow sphere is spinning around the bucket, or the bucket is spinning within the hollow sphere. Just as Mach advocated, the only thing that matters is the relative spinning motion between the two. And since the calculations I’ve referred to make use of nothing but general relativity, this is an explicit example of a distinctly Machian feature of Einstein’s theory. (Nevertheless, whereas standard Machian reasoning would claim that the water would stay flat if the bucket spun in an infinite, empty universe, general relativity disagrees. What the Pfister and Braun results show is that a sufficiently massive rotating sphere is able to completely block the usual influence of the space that lies beyond the sphere itself.)